Run report: MutantXL closure at degree 4, N = 26–42

Study `research/dreg_closure_scale_20261016/`, pre-registered in `PREREGISTRATION.md` (commit `f10c86d38`). **Run** 2026-10-16, 17:05 to 18:09 UTC. **Class:** stage diagnostic, with no `S`, no rho ratio and no speed claim.

What ran

- **Draws.** `run.sh LADDER CLOSURE gen` generated the draws with seed 20261016 (exact solution count; XL built to degree 3 only), and they were committed before any measurement.
- **Measurement.** `run.sh … measure` ran one process per unsatisfiable draw, as `prlimit --as=15e9 timeout 7200 dreg_closure --to 4 --max-cols 130000 --draw K`. `run.sh … control` ran the soundness control.
- **Binaries.** SHA-256 values are in `runs/binary.sha256`.
- **Cells.**
  - Committed m = 3 ℓ = 5 draws: (11,5), (13,5) and (15,5), N = 26–30, XL ≥ 7.
  - Fresh m = 3 draws: (16,6), (18,6), (19,7) and (21,7).
  - Fresh m = 4 draws: (12,3), (14,3) and (13,4).
  - Four unsatisfiable draws per cell, 40 in total.
- **Host.** The 4-core, 16 GB container, which is not isolated (`AGENTS.md` §10). Wall times are practicality notes only.

Completion

- **Processes.** 41 of 41 (40 draws plus 1 control) exited 0. None was killed by the memory or time limit.
- **Degree 4.** It was built in full on every draw, and every closure saturated.
- **Peak RSS.** 7.1 GB at (13,4) m = 4. The slowest draw took 283 s.

Measured values

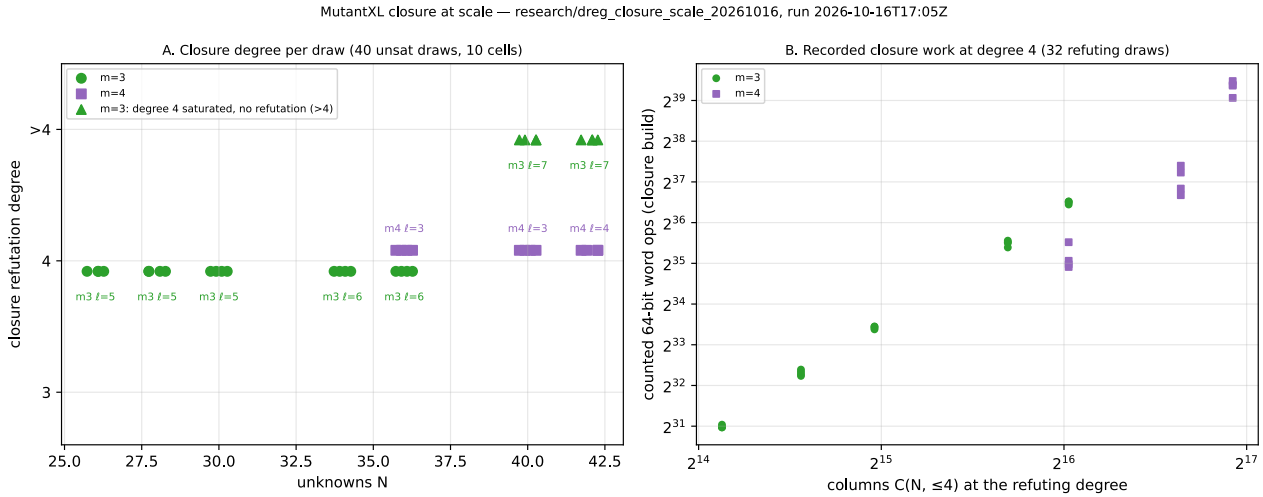


Figure 1: Recorded values from `runs/*.d*.jsonl` (40 draws, run 2026-10-16T17:05Z). A: the closure refutation degree per draw against the unknowns N. Triangles mark draws whose degree-4 closure saturated without refuting. B: the counted closure word operations against the columns $C(N, \leq 4)$ at the refuting degree. Source: `plot.py` (layout only).

- **Closure degree.** 4 on all 32 draws of eight cells.
- **Above 4.** All 8 draws of m = 3 ℓ = 7 (N = 40, 42).
- **XL.** It refuted at degree 4 on none of the 40 draws.
- **Work ratio.** $\log_2$(word ops) ÷ $\log_2$(columns) was 2.18–2.33 on the refuting draws.
- **Per-draw values.** In `score-output.txt` (native `dreg_score scale`).

Control

Soundness, as registered: the first two satisfiable draws of each fresh cell with $N \leq 36$. Only one exists, `m4-12-3` draw 0, which has 24 solutions. Built to degree 4, it was refuted by neither XL nor the closure. Records: `runs/control-m4-12-3.d0.{jsonl,log}`.

Raw records

- `runs/*.d*.jsonl`: one line per draw, with every degree's XL and closure ranks, columns, products, word ops and seconds.
- `runs/*.log`: peak VmHWM per process.

- `runs/queue.log`; `draws/*.jsonl` (generator output).
- The interpretation is in `../../RESULTS.md`.